

Sunday 28th of June, 8:35 to 12:00



Time	Description	Discussant
[15 min]	(0) Introductions and administrivia	Trikalinos
[25 min]	(1) DES as a composition of point processes	Alarid-Escudero
[30 min]	(2) NHPPPs – key properties	Trikalinos
[30 min]	(3) Sampling from NHPPPs	Trikalinos
[15 min]	Break	
[80 min]	(4) Working with code -- Base R vs <code>nhppp</code> : Simulate first vs all events in $(a, b]$ -- Packaging a simple cancer DES -- The vectorized case	[All] Trikalinos Sereda Sereda
[5 min]	(5) Advanced Topic Teaser on self-excitatory processes: point processes that are not NHPPPs and when you may need them	Trikalinos
[15 min]	General Q & A	All

Section 3: Sampling

Three important properties for sampling

Memorylessness

You can ignore what happens outside your interval

Composability

You can merge two NHPPs with intensities λ_1, λ_2 to get a new NHPP with intensity $\lambda_1 + \lambda_2$.

Transmutability (time warping)

Any one-to-one transformation of the intensity function results in a unique NHPP in the transformed time axis

Overview of the sampling strategy

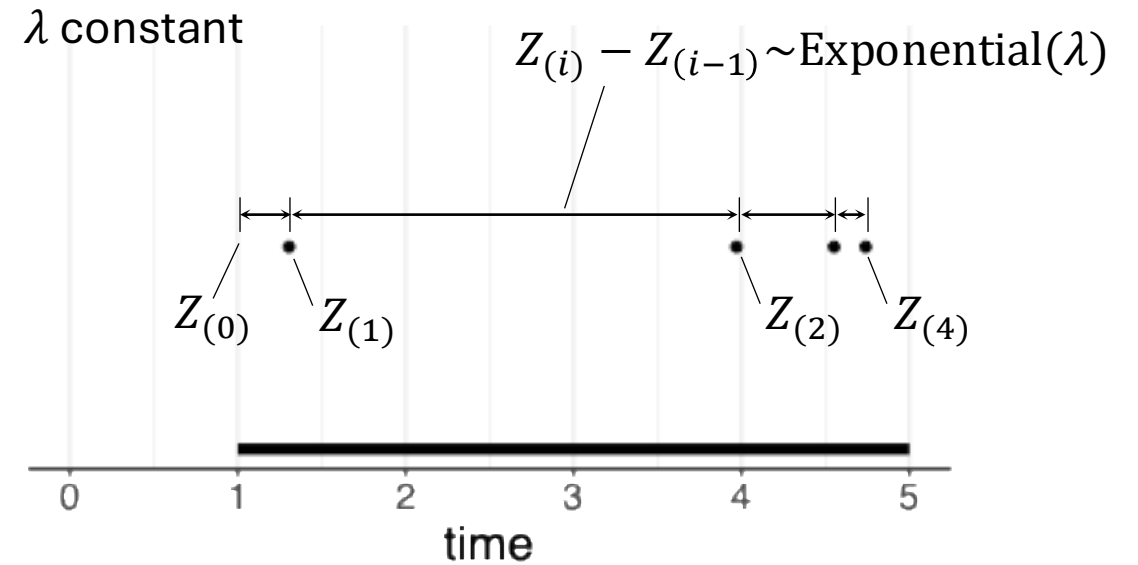
1. Sampling from constant rate PPP is easy *easy*
2. Memorylessness implies you can treat the piecewise as constant PPPs over disjoint interval *peasy*
3. Composability motivates an acceptance-rejection algorithm for sampling from any $\lambda(t)$ *almost always practical*
4. Time warping allows efficient sampling if you have (cheap access to) $\Lambda(t), \Lambda^{-1}(t)$ *sometimes possible, may be worth the hassle to get Λ, Λ^{-1}*

1. Sampling from a PPP is easy

Constant intensity function (homogeneous PPP)

Sampling from a constant intensity function is easy.

The interarrival times have an exponential distribution.

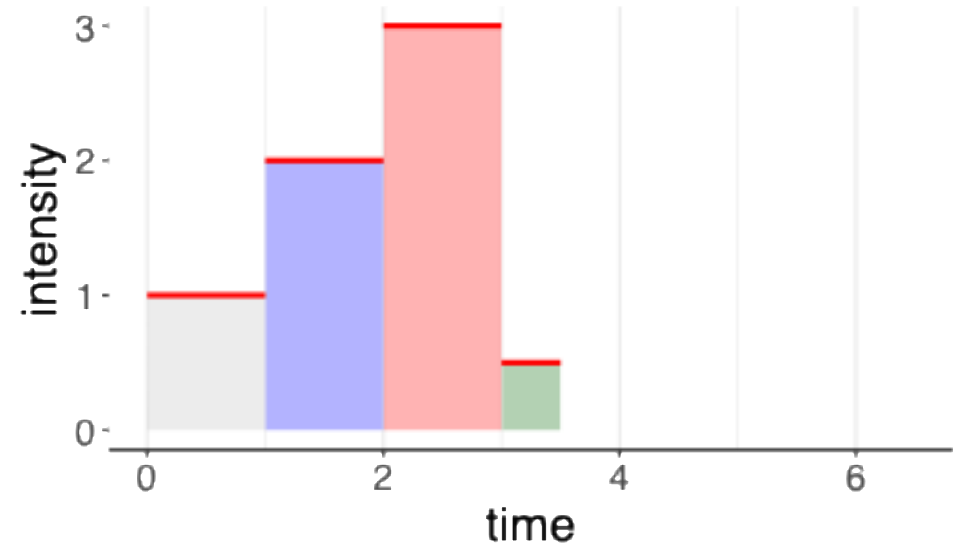


2. Memorylessness: Sampling
from piecewise constant NHPPP
is peasy

Piecewise constant intensity function (NHPPP)

- Look at each piecewise constant interval separately
- In each interval you have a constant intensity (easy)
- Return the union of all events

Sampling from piecewise constant intensities is easy (**memorylessness**)



3. Composability: Sampling NHPPPs when you know $\lambda(t)$ reduces to sampling from a PPP (#1) or piecewise constant NHPPP (#2)*

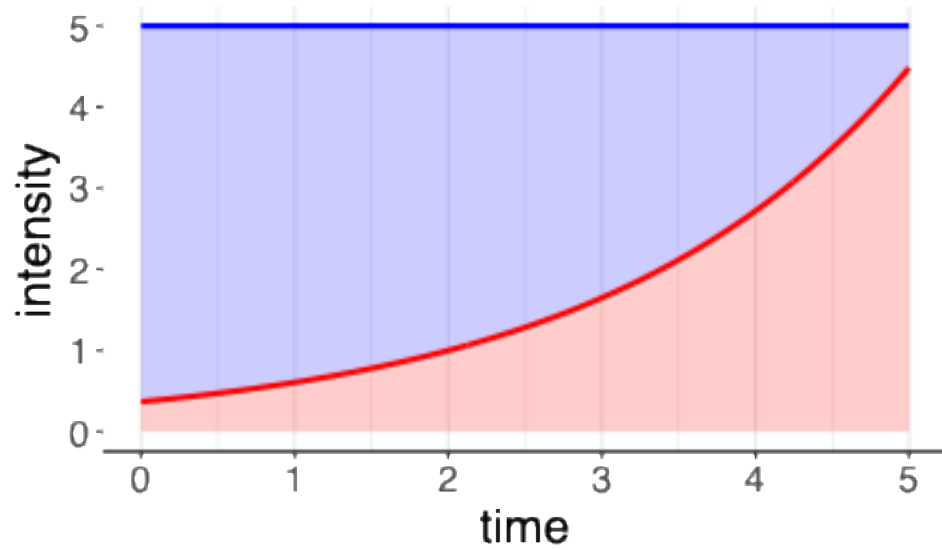
** You still need to find a constant or piecewise constant majorizer $\lambda_*(t)$, whose choice determines your efficiency .*

You cannot get achieve something difficult with zero effort.

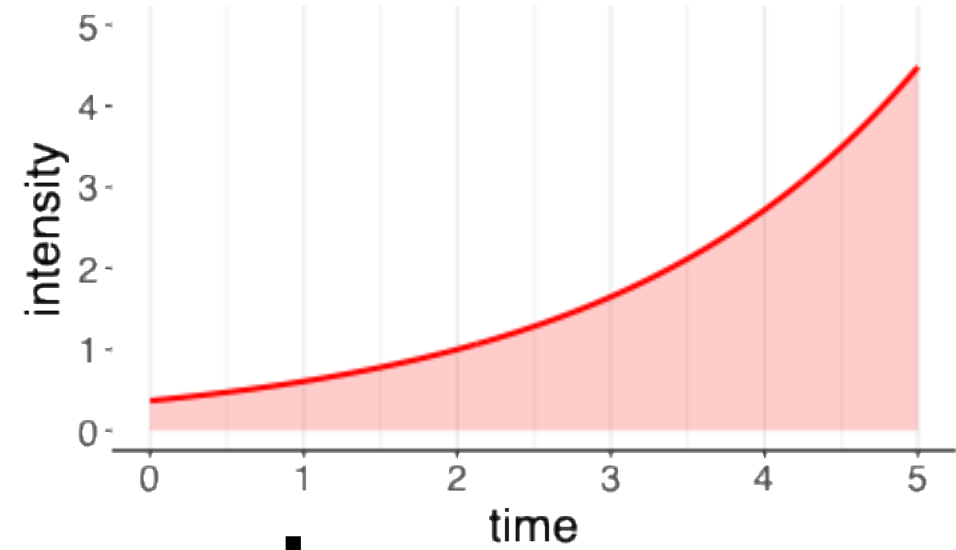
You will put in some work.

Other terms and conditions may apply.

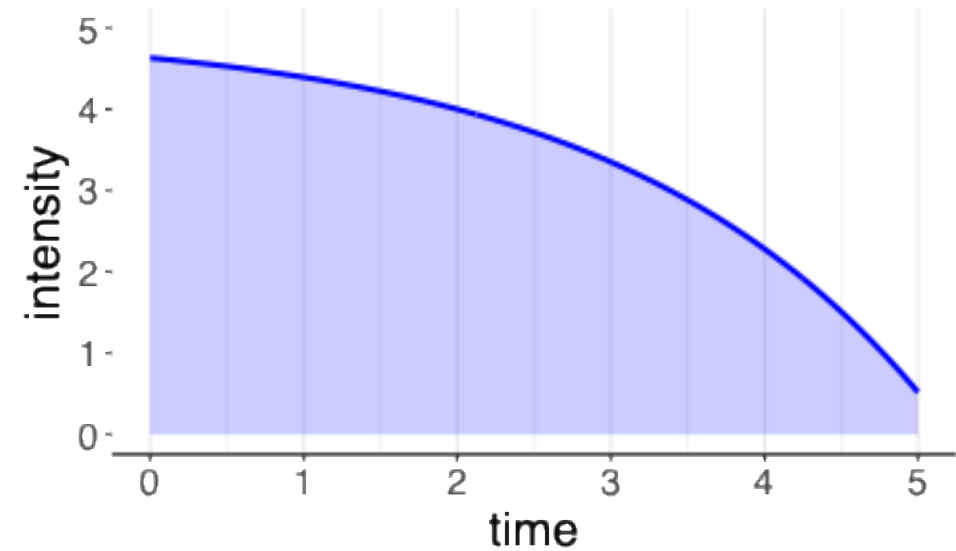
Composability



=

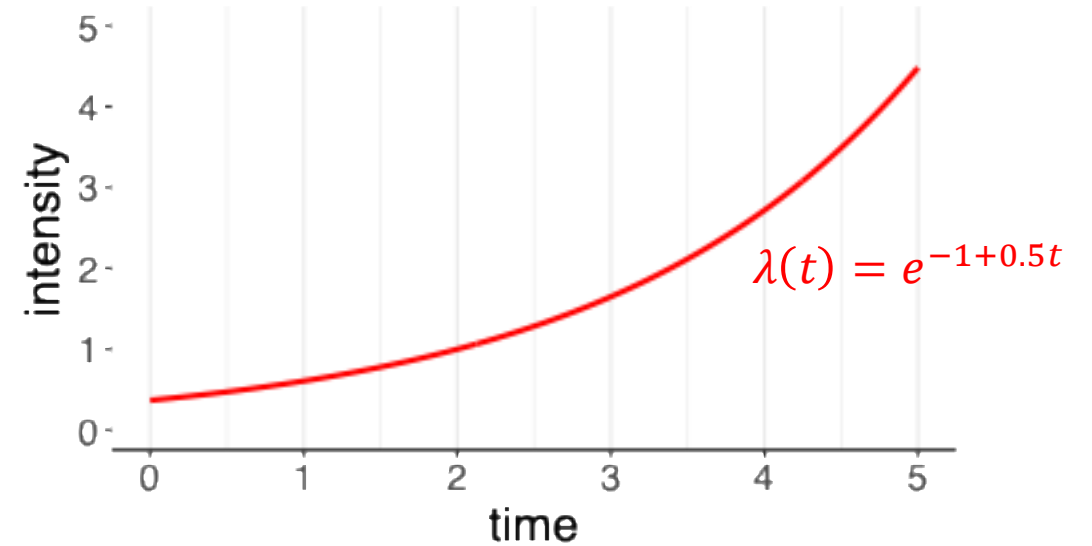


+



NHPPP, where you know $\lambda(t)$: Thinning

The general case is more challenging



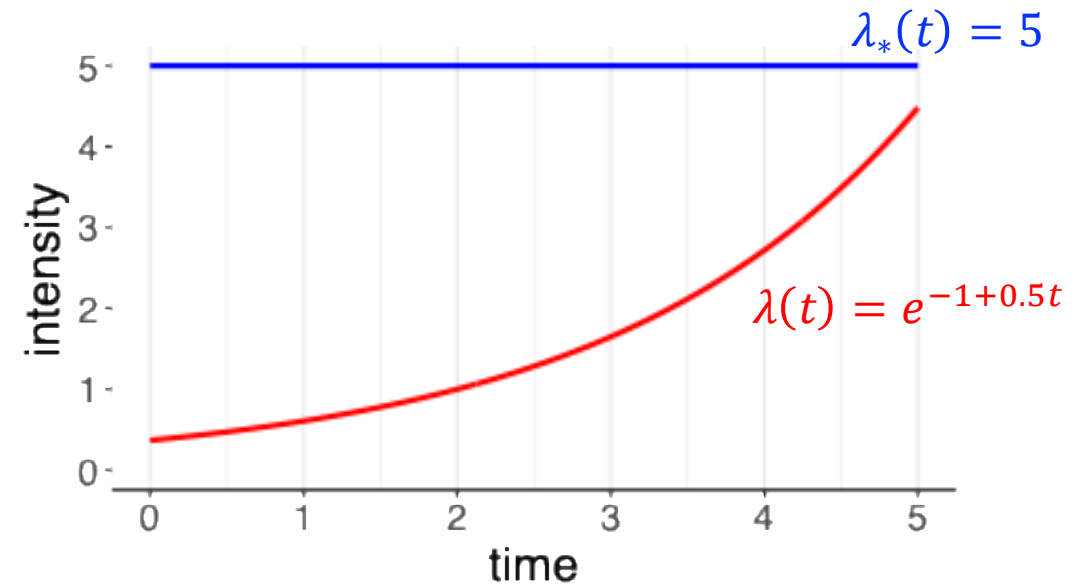
NHPPP, where you know $\lambda(t)$: Thinning

- Find a majorizer function λ_* that's easy to sample

Majorizer: any function that is “taller” than λ

$$\lambda_* \geq \lambda$$

(and has the same support as λ)



NHPPP, where you know $\lambda(t)$: Thinning

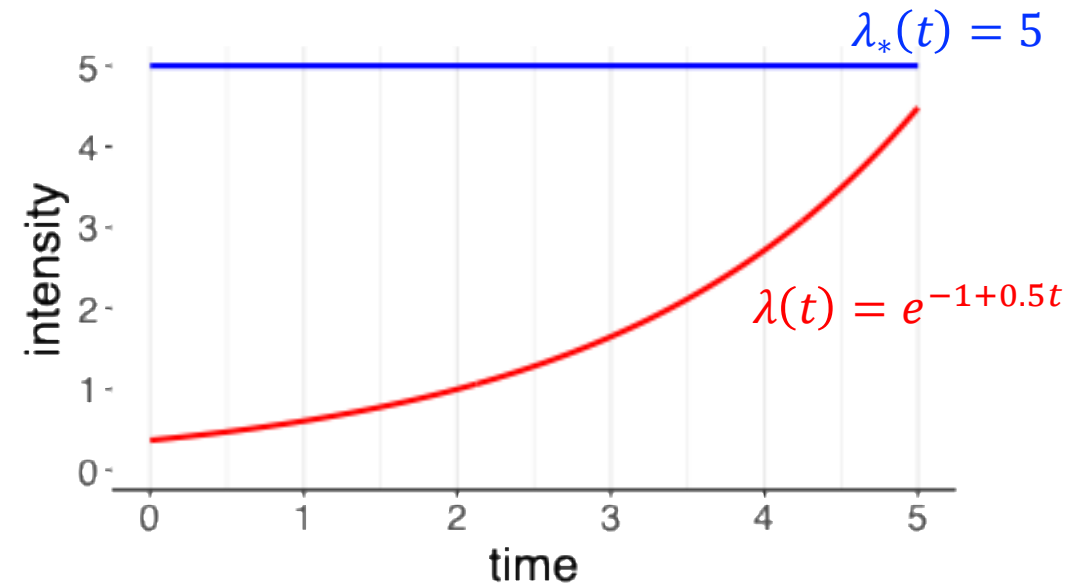
- Find a majorizer function λ_* that's easy to sample

$$\lambda_*(t) = \lambda(t) + [\lambda_*(t) - \lambda(t)]$$

Sample proposals from here

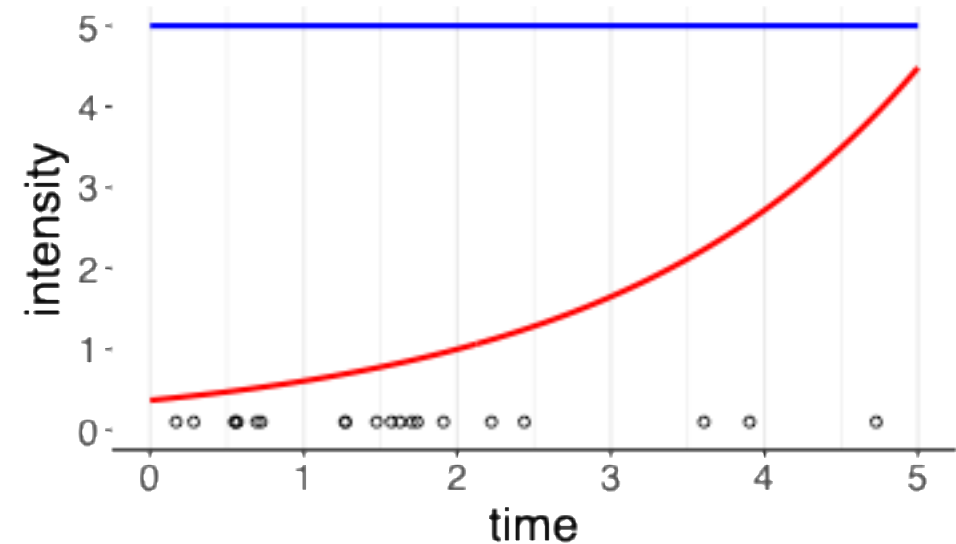
Keep proposals conforming to this part

Reject the rest



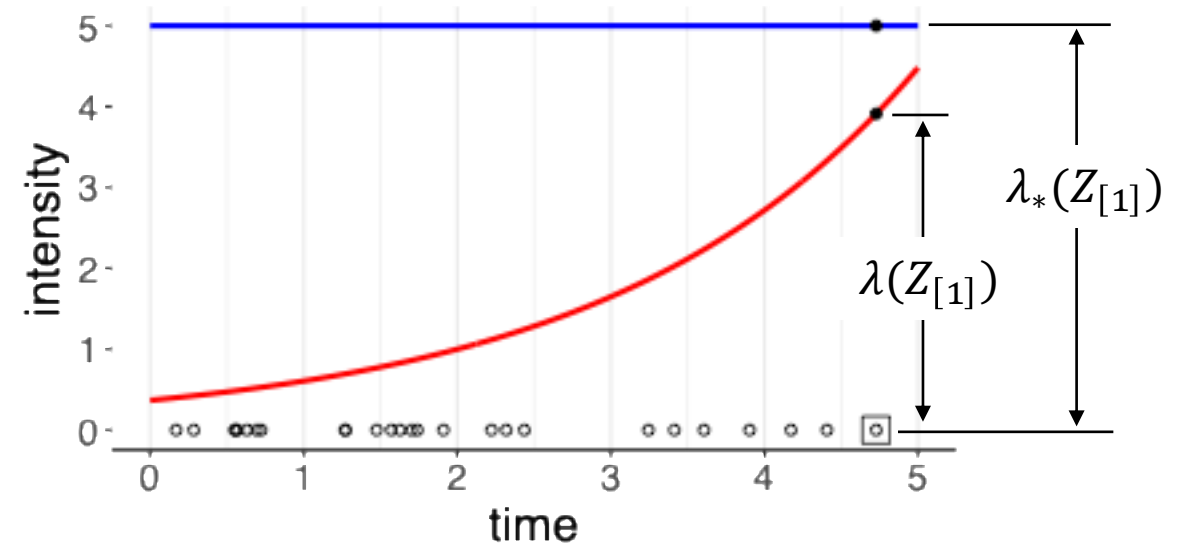
NHPPP, where you know $\lambda(t)$: Thinning

- Find a majorizer function λ_* that's easy to sample
- Draw events $\{Z_{*1}, \dots\}$ from λ_*



NHPPP, where you know $\lambda(t)$: Thinning

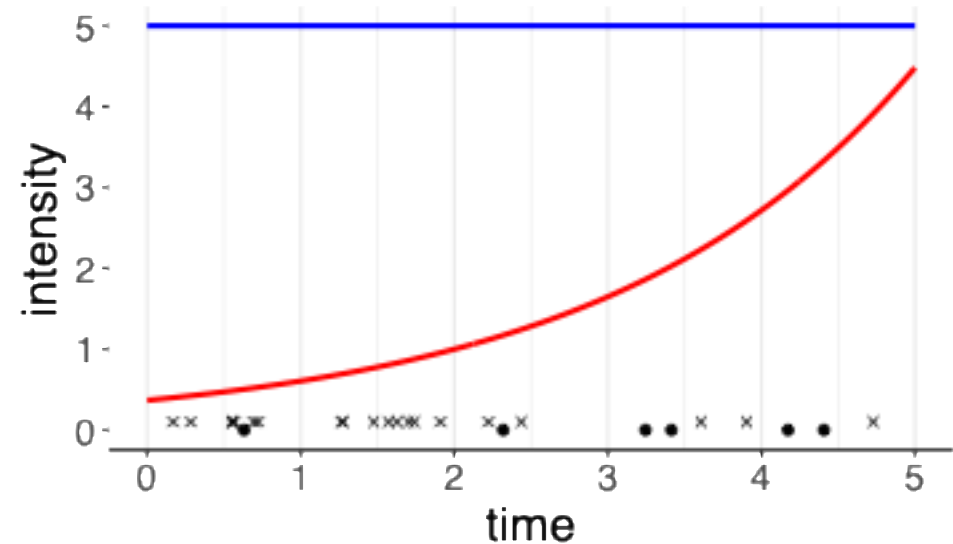
- Find a majorizer function λ_* that's easy to sample
- Draw events $\{Z_1, \dots\}$ from λ_*
- Accept event i with probability $\frac{\lambda(Z_i)}{\lambda_*(Z_i)}$



NHPPP, where you know $\lambda(t)$: Thinning

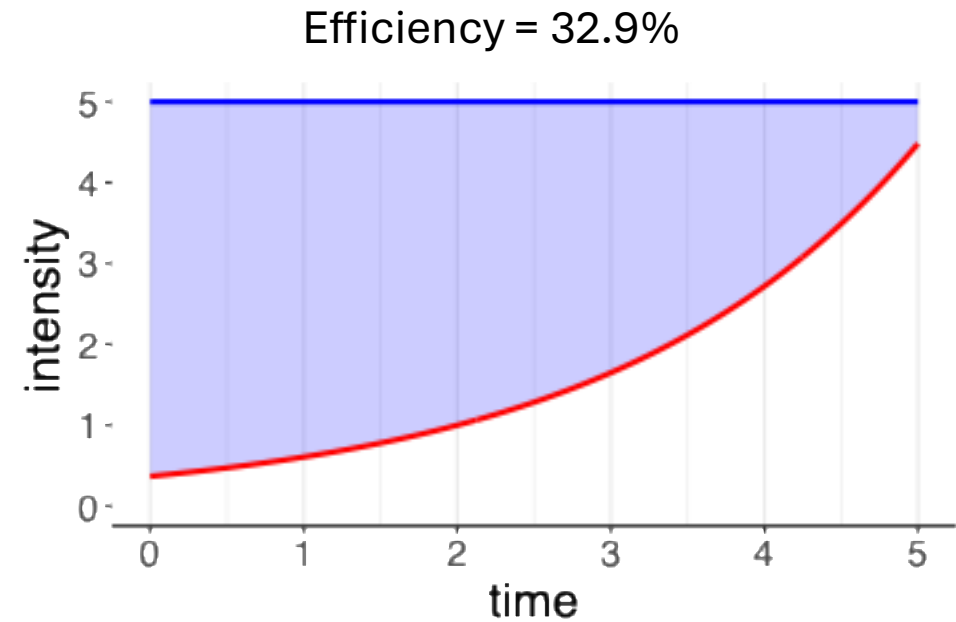
- Find a majorizer function λ_* that's easy to sample
- Draw events $\{Z_{*1}, \dots\}$ from λ_*
- Accept event i with probability $\frac{\lambda(Z_i)}{\lambda_*(Z_i)}$
- The set of accepted points is an instantiation from $\lambda(t)$

(composability)



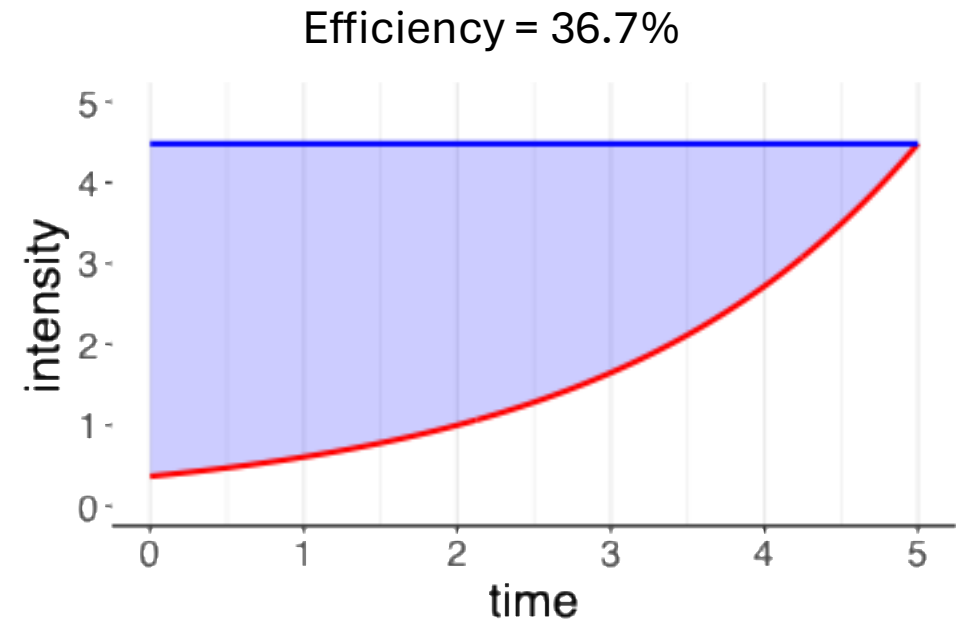
Thinning, efficiency

- Thinning efficiency: average fraction of proposals that are accepted
- Depends on the choice of λ_*
- The smaller the blue area, the better the efficiency



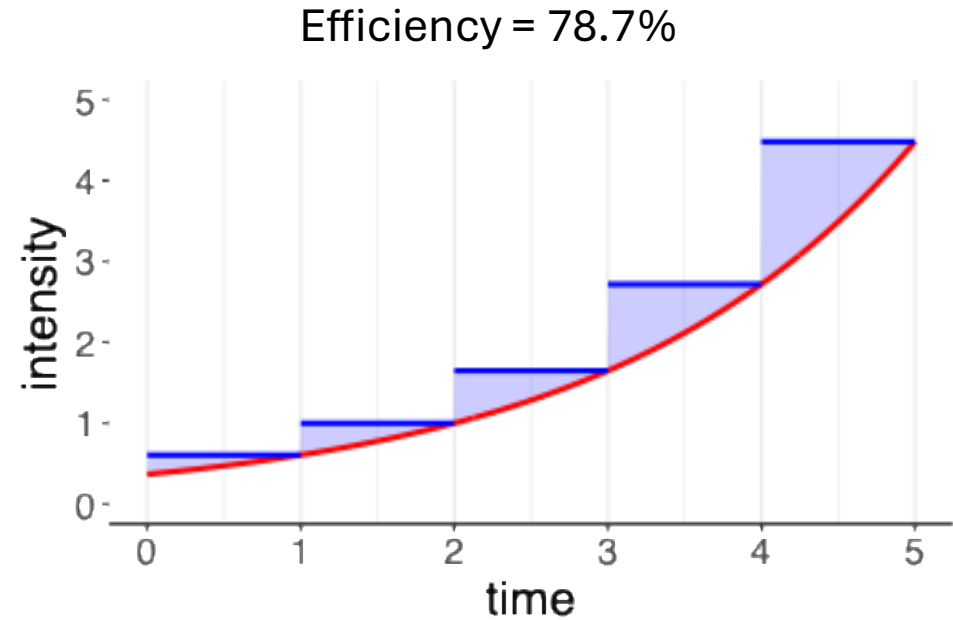
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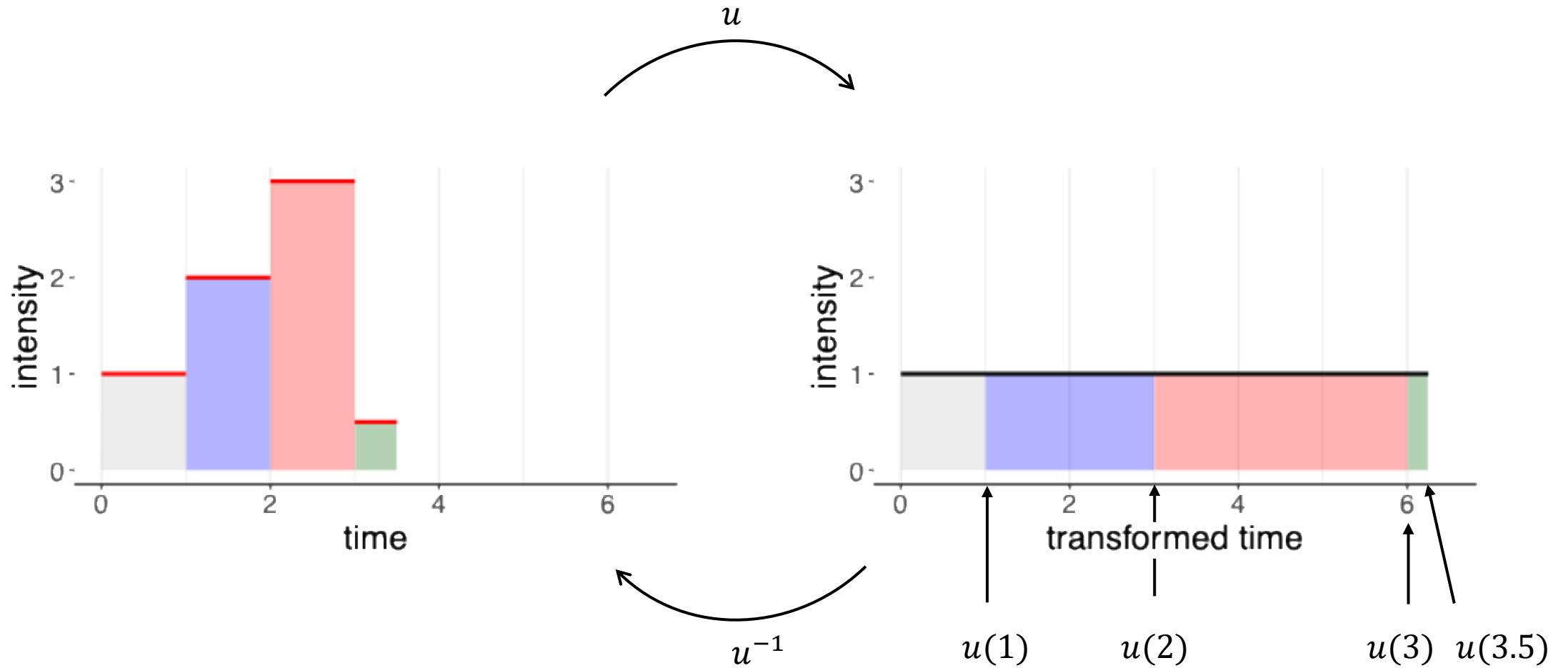


4. Transmutability of time: Sampling NHPPPs when you know Λ, Λ^{-1} reduces to sampling from a PPP with rate one (#1) *

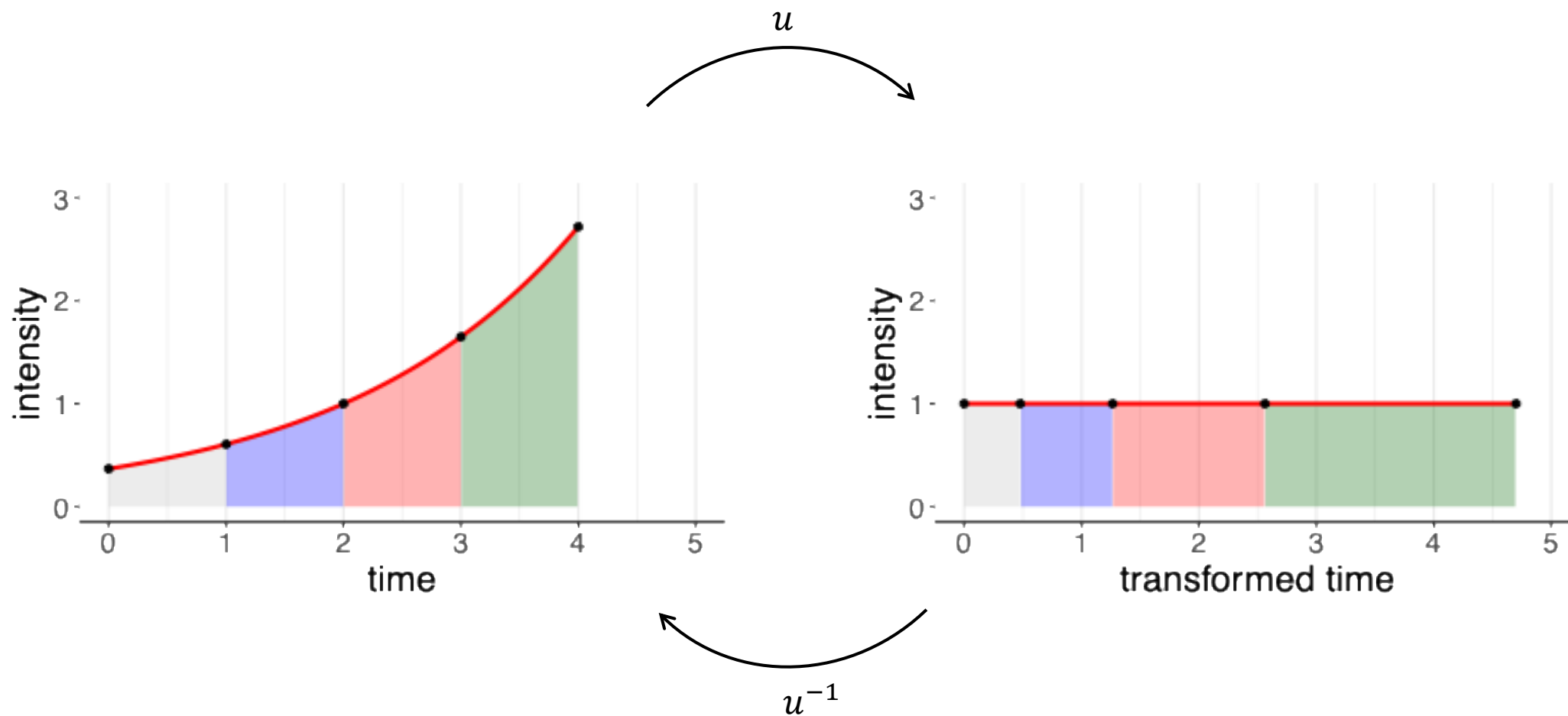
** You will need to do some maths to get Λ, Λ^{-1} . It may not be practical to do so, or even possible. In such a case, back to (#3). Even if you have Λ , you may not have a cheap Λ^{-1} .*

You cannot achieve something difficult with zero effort. You will put in some work. Other terms and conditions may apply.

Transmutability



Transmutability



A nice u is Λ (and then u^{-1} is Λ^{-1})

Change of variable from s to u

$$\Lambda(t) = \int_a^t \lambda(s) \, ds = \int_{u(a)}^{u(t)} \frac{\lambda(s)}{u'(s)} \, du$$

Pick u so that $u' = \lambda$. Any antiderivative of λ works. Using $u := \Lambda$, transforms time to scale where the process has constant rate 1,

$$\int_{\Lambda(a)}^{\Lambda(t)} \frac{\lambda(s)}{\Lambda'(s)} \, du = \int_{\Lambda(a)}^{\Lambda(t)} 1 \, du .$$

This is a sketch of the formal proof – omitting the rigorous bits

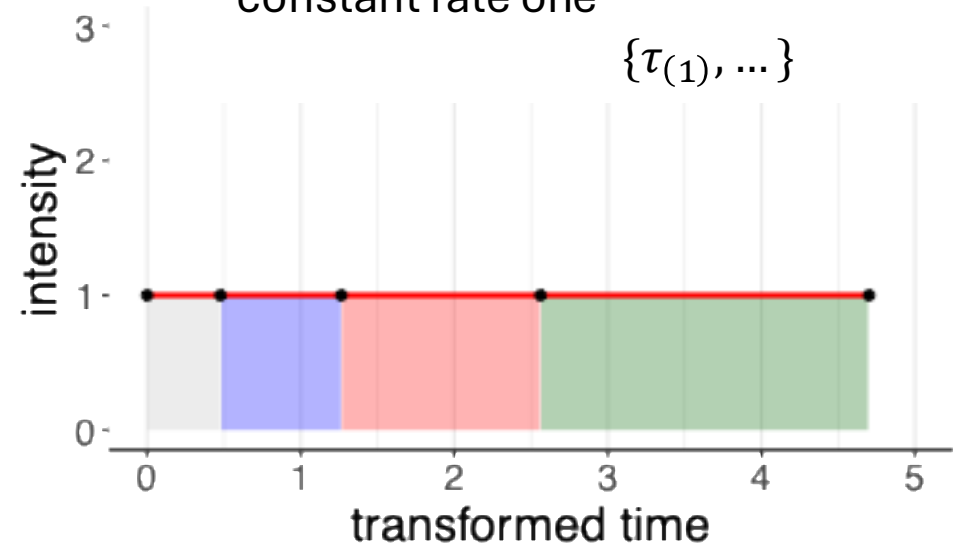
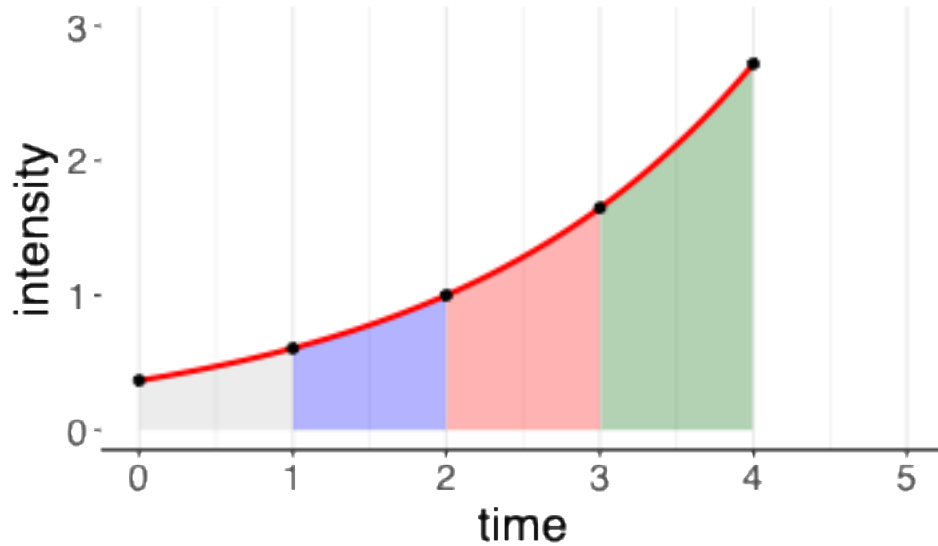
Transmutability

1. Find the start and stop of the transformed time interval
 $\tau_{start} = \Lambda(t_{start})$ and $\tau_{stop} = \Lambda(t_{stop})$

$$\tau_{start} = \Lambda(t_{start}) \text{ and } \tau_{stop} = \Lambda(t_{stop})$$

Λ

2. Sample transformed times from a PPP with constant rate one
 $\{\tau_{(1)}, \dots\}$



3. Back-transform the instantiation to the original time scale
 $\{\Lambda^{-1}(\tau_{(1)}), \dots\}$

$$\{\Lambda^{-1}(\tau_{(1)}), \dots\}$$

Λ^{-1}

More in these works...

PLOS ONE

RESEARCH ARTICLE

The nhppp package for simulating non-homogeneous Poisson point processes in R

Thomas A. Trikalinos^{1,2,3*}, Yuliia Sereda¹

Original Research Article

A Fast Nonparametric Sampling Method for Time to Event in Individual-Level Simulation Models

David U. Garibay-Treviño¹, Hawre Jalal¹, and Fernando Alarid-Escudero¹

MDM
Medical Decision Making

Medical Decision Making
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Next ... Section 4: Hands-on
example (simple case)

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